

ENVS 193DS Midterm

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Set up

```
# loading in packages
library(tidyverse)
library(janitor)
library(gt)
library(rstatix)
# reading in data
halibut <- read_csv("halibut.csv")
bee_pathogens <- read_csv("bee-pathogens.csv")
```

Problem 1

(a) Test type

This test could have been a two-sample t-test or a Mann-Whitney U-test. These would work because they both compare two different populations. However, they are different because the Mann-Whitney U-test works best when the median is a better measure of central tendency, and a two-sample t-test works better when the mean is the better measure of central tendency.

(b) Writing new statements

Two-sample t-test: Across years (1969-2019), we found no difference in commercial halibut landings between Northern ($n =$, $\text{mean} =$) and Southern ($n =$, $\text{mean} =$) California (____
two-sample t-test: $t(df) = t$, $p = 0.3$, $\alpha =$).

Mann-Whitney U-test: Across years (1969-2019), we found no difference in commercial halibut landings between Northern (median = , n =) and Southern (median = , n =) California (Mann-Whitney U-test, $W =$, $p = 0.3$, $\alpha =$).

(c) New figures

```
# creating new object from halibut data frame
halibut_clean <- halibut |>
  # making data more readable by making the units 100,000lbs
  mutate(landings_100k_lb = landings_lb/100000)
# removing landings_lb column
halibut_clean$landings_lb <- NULL
```

```
# doing F test to determine variance
var.test(landings_100k_lb ~ ca_region,
  # data: halibut data frame
  data = halibut_clean)
```

F test to compare two variances

```
data:  landings_100k_lb by ca_region
F = 1.2599, num df = 50, denom df = 50, p-value = 0.4169
alternative hypothesis: true ratio of variances is not equal to 1
95 percent confidence interval:
 0.7191383 2.2072803
sample estimates:
ratio of variances
      1.259897
```

This test shows that the variance is approximately equal (F-test, $F(50, 50) = 1.25$, $p = 0.42$).

```
# doing t-test to compare regions
t.test(
  landings_100k_lb ~ ca_region,
  # variances are equal
  var.equal = TRUE,
  # data is from halibut data frame
  data = halibut_clean)
```

Two Sample t-test

```
data: landings_100k_lb by ca_region
t = 1.0479, df = 100, p-value = 0.2972
alternative hypothesis: true difference in means between group Northern and group Southern is
95 percent confidence interval:
 -0.356636  1.155190
sample estimates:
mean in group Northern mean in group Southern
      3.906430           3.507153
```

```
# creating new object from halibut_clean
halibut_ci <- halibut_clean |>
  # grouping by region
  group_by(ca_region) |>
  summarize(
    # mean
    mean = mean(landings_100k_lb),
    # n
    n = length(landings_100k_lb),
    # degrees of freedom
    df = n - 1,
    # standard deviation
    sd = sd(landings_100k_lb),
    # standard error
    se = sd/sqrt(n),
    # t value
    tval = qt(p = 0.05/2, df = df, lower.tail = FALSE),
    # margin of error
    margin = tval*se,
    # lower bound of the CI
    ci_lower = mean - margin,
    # upper bound of the CI
    ci_higher = mean + margin
  ) |>
  # selecting values to display in data frame
  select(ca_region, mean, ci_lower, ci_higher)
```

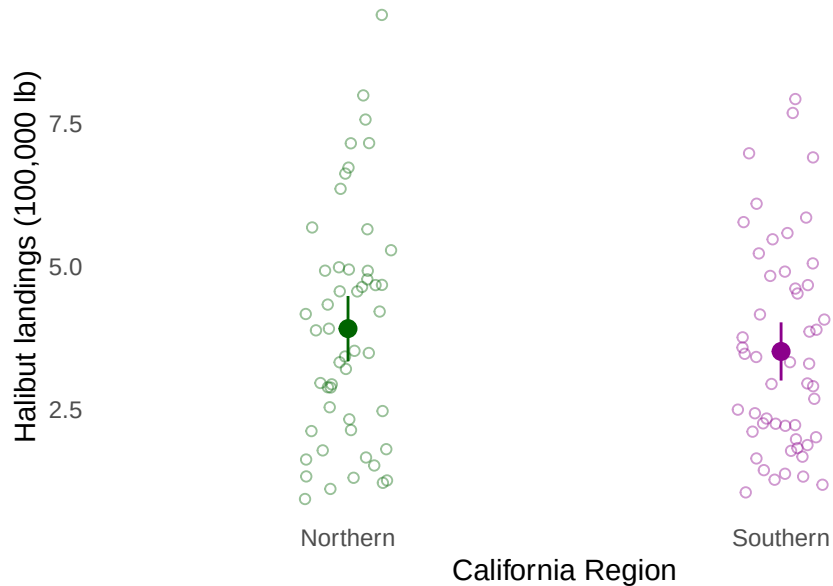
```
# base ggplot from halibut data frame
ggplot(data = halibut_clean,
  # x-axis: region
```

```

    mapping = aes(x = ca_region,
                  # y-axis: landings
                  y = landings_100k_lb,
                  # coloring by region
                  color = ca_region)) +
# first layer: jitterplot
# no jitter in vertical direction
geom_jitter(height = 0,
            # smaller jitter in horizontal direction
            width = 0.1,
            # making points transparent
            alpha = 0.4,
            # making points circles
            shape = 21) +
# second layer: showing means with 95% CI
geom_pointrange(data = halibut_ci,
               aes(x = ca_region,
                   y = mean,
                   ymax = ci_lower,
                   ymin = ci_higher)) +
# labeling the axes
labs(x = "California Region",
     y = "Halibut landings (100,000 lb)",
     # adding a title
     title = "Higher mean halibut landings in Northern California") +
# changing colors
scale_color_manual(values = c("Northern" = "darkgreen",
                              "Southern" = "magenta4")) +
# removing legend
theme(legend.position = "none",
      # making background blank
      panel.background = element_blank(),
      panel.grid = element_blank(),
      axis.ticks = element_blank())

```

Higher mean halibut landings in Northern California



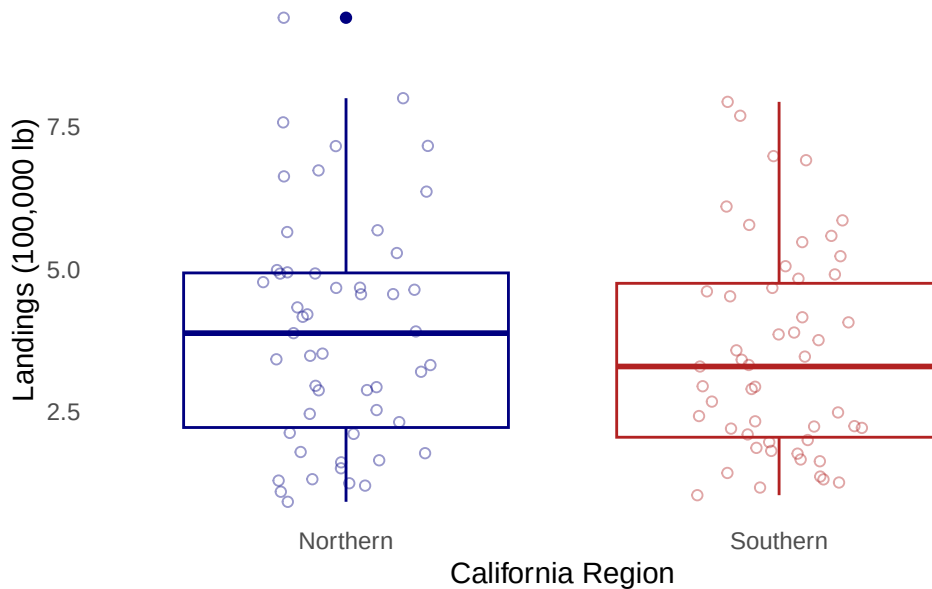
```
# base ggplot from halibut_clean data frame
ggplot(data = halibut_clean,
  # x-axis: region
  mapping = aes(x = ca_region,
    # y-axis: landings
    y = landings_100k_lb,
    # color by region
    color = ca_region)) +
# first layer: boxplot
geom_boxplot() +
# second layer: jitterplot
geom_jitter(
  # no jitter in vertical direction
  height = 0,
  # smaller jitter in horizontal direction
  width = 0.2,
  # making points transparent
  alpha = 0.4,
  # making points circles
  shape = 21) +
# adding axis labels
labs(x = "California Region",
  y = "Landings (100,000 lb)",
```

```

# adding title
title = "Median halibut landings higher in Northern California ") +
# changing colors
scale_color_manual(values = c("Northern" = "navyblue",
                              "Southern" = "firebrick"))+
# removing legend
theme(legend.position = "none",
      # making background blank
      panel.background = element_blank(),
      panel.grid = element_blank(),
      axis.ticks = element_blank())

```

Median halibut landings higher in Northern California



(d) Caption

Figure 1. Difference between mean halibut landings in Northern and Southern California. Points represent yearly halibut landings (100,000 lbs) in Northern (shown in green) or Southern (shown in purple) California. The center filled point represents the mean, and the bars represent a 95% confidence interval. Data from California Department of Fish and Wildlife. 2022. California halibut, *Paralichthys californicus*, Enhanced Status Report. Data accessed 2026-05-04

Figure 2. Difference between median halibut landings in Northern and Southern California. The left (blue) boxplot shows halibut landings (100,000 lbs) in Northern California, and the

right (red) boxplot shows landings in Southern California. The boxes represent the interquartile range of the landings in each region. The center bold bar in the box represents the median of each region. The whiskers extend to the minimum and maximum values (excluding outliers). The circles represent yearly halibut landings in each region. Data from California Department of Fish and Wildlife. 2022. California halibut, *Paralichthys californicus*, Enhanced Status Report. Data accessed 2026-05-04

Problem 2

(a) Plan your figure

x-axis: year

y-axis: landings_lb

geometry function: geom_line

colors and linetypes: ca_region

relabelling the x- and y-axes: labs()

using custom colors: scale_color_manual()

theme arguments and the element_ for: positioning the x-axis text at a 45° angle:
theme(axis.text.x = element_text(angle = 45))

putting the legend in the top right of the plot: theme(legend.position = c(0.95, 0.95),
legend.justification = c("right", "top"))

removing the panel background: theme(panel.background = element_blank())

removing the panel grid: theme(panel.grid = element_blank())

removing the axis ticks: theme(axis.ticks = element_blank())

removing the legend title: legend.title = element_blank()

(b) Code your figure

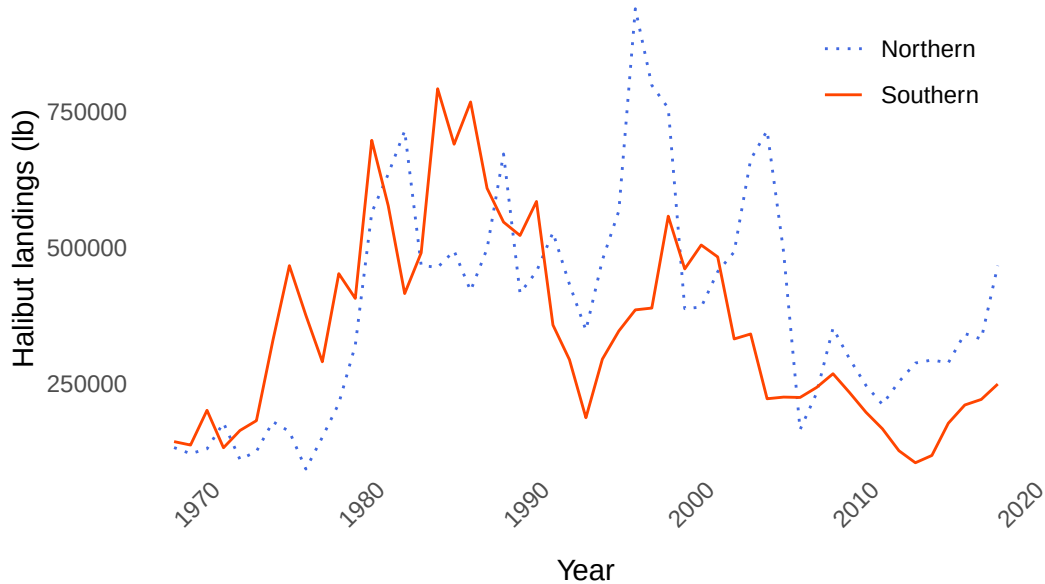
```
# base layer: ggplot
# data from halibut
ggplot(data = halibut,
       # x-axis: year
       aes(x = year,
          # y-axis: landings
```

```

    y = landings_lb,
    # color by region
    color = ca_region,
    # change line type by region
    linetype = ca_region)) +
# first layer: geom_line
geom_line() +
# manually changing linetype
scale_linetype_manual(values = c(
  "Northern" = "dotted",
  "Southern" = "solid")) +
# add axis titles and subtitle
labs(x = "Year",
     y = "Halibut landings (lb)",
     subtitle = "Lucia Murillo") +
theme(
  # make x-axis labels at a 45 degree angle
  axis.text.x = element_text(angle = 45),
  # move legend to top right in plot
  legend.position = c(0.95, 0.95),
  legend.justification = c("right", "top"),
  # make background blank
  panel.background = element_blank(),
  panel.grid = element_blank(),
  axis.ticks = element_blank(),
  # remove legend title
  legend.title = element_blank()) +
# changing colors
scale_color_manual(values = c("Northern" = "royalblue",
                              "Southern" = "orangered"))

```


Lucia Murillo



(c) Caption

Figure 3. Halibut landings in Northern and Southern California over time. Solid blue line represents Northern California halibut landings (lbs) over time, and the dotted orange line represents halibut landings in Southern California over time. Data from California Department of Fish and Wildlife. 2022. California halibut, *Paralichthys californicus*, Enhanced Status Report. Data accessed 2026-05-04.

Problem 3

(a) Variables

The response variable for this experiment is black queen cell virus (BQCV) loads in *B. impatiens*, and it is a continuous variable. The predictor variable is the habitat type (edge, forest, valley). BQCV loads were measured by performing quantitative PCR tests of the pooled extracts of the *B. impatiens* after extracting the RNA (Methods: Pathogen Quantification section in paragraph 1-2).

(b) Test type

These results were obtained by doing a Kruskal-Wallis test followed by a Dunn post-hoc test among habitat types using a Bonferroni correction to adjust the P-value for multiple comparisons (Statistical analysis paragraph).

(c) Hypotheses

Null hypothesis: The median BQCV load in *B. impatiens* is the same in all habitat types.
Alternative hypothesis: The median BCQV load in *B. impatiens* is different in different habitat types.

(d) Cleaning and organizing data

```
# creating new object from bee_pathogens
bee_pathogens_clean <- bee_pathogens |>
  # cleaning names
  clean_names() |>
  # selecting site_type_x and bqc
  select(site_type_x, bqc) |>
  # renaming site_type_x to habitat
  rename(habitat = site_type_x) |>
  # removing N/A values
  drop_na()
#displaying 10 rows from bee_pathogens_clean
bee_pathogens_clean |>
  slice_sample(n = 10)
```

	habitat	bqc
1	forest	5.092574
2	valley	5.108063
3	forest	4.997318
4	edge	2.440369
5	forest	1.098139
6	edge	6.146976
7	valley	10.008824
8	edge	8.409579
9	edge	3.495354
10	forest	1.912831

Bee Pathogen Summary

Habitat type	Median BQCV	Sample size
edge	6.3	27.0
forest	5.0	24.0
valley	7.6	29.0

(e) Summarizing and table displays

```
# creating a new object from clean dataframe
bee_pathogens_summary <- bee_pathogens_clean |>
  # grouping by habitat
  group_by(habitat) |>
  # calculating mean and sample size
  summarize(median_bqcv = median(bqcv),
            n = length(bqcv)) |>
  # ungrouping data frame
  ungroup()
# creating table
bee_pathogens_summary |>
  gt() |>
  # rounding decimals
  fmt_number(decimals = 1) |>
  # changing column names
  cols_label(habitat = "Habitat type",
             median_bqcv = "Median BQCV",
             n = "Sample size") |>
  # changing header for table
  tab_header(title = "Bee Pathogen Summary")
```

(f) Test

```
# running kruskal-wallis test
# BQCV by habitat
kruskal.test(bqcv ~ habitat, data =
  # data from bee_pathogens_clean
  bee_pathogens_clean)
```

Kruskal-Wallis rank sum test

data: bqcv by habitat

Kruskal-Wallis chi-squared = 9.8219, df = 2, p-value = 0.007365

(g) Post-hoc test

```
# running dunn post hoc test
# median BQCV by habitat
dunn_test(bqcv ~ habitat,
          # data from bee_pathogens_summary
          data = bee_pathogens_clean,
          # bonferroni correction
          p.adjust.method = "bonferroni")
```

A tibble: 3 x 9

	.y.	group1	group2	n1	n2	statistic	p	p.adj	p.adj.signif
*	<chr>	<chr>	<chr>	<int>	<int>	<dbl>	<dbl>	<dbl>	<chr>
1	bqcv	edge	forest	27	24	-1.83	0.0679	0.204	ns
2	bqcv	edge	valley	27	29	1.31	0.189	0.567	ns
3	bqcv	forest	valley	24	29	3.13	0.00175	0.00525	**

(h) Effect size

```
# dataframe: bee_pathogens_clean
bee_pathogens_clean |>
  # calculating kruskal-wallis effect size
  # median BQCV by habitat
  kruskal_effsize(bqcv ~ habitat)
```

A tibble: 1 x 5

	.y.	n	effsize	method	magnitude
*	<chr>	<int>	<dbl>	<chr>	<ord>
1	bqcv	80	0.102	eta2[H]	moderate

(i) Written communication

We found a moderate ($\eta^2 = 0.102$, $n = 80$) difference in median BQCV loads between edge, forest, and valley habitats (Kruskal-Wallis test $\chi^2(2) = 9.82$, $p = 0.007$, $\alpha = 0.05$). Bees in forest habitats had the lowest BQCV loads (median = 4.59, $n = 24$), which differed from valley habitats (median = 7.48, $n = 29$; Dunn's post-hoc test: adjusted p-value = 0.005) but edge habitats (median = 6.32, $n = 27$) were not different from either forest (adjusted p-value = 0.204) or valley habitats (adjusted p-value = 0.567).